

3.5 Systems of Linear Equations [Recall - Gaussian Elimination]

- Gaussian elimination is a direct method for solving systems of linear equations

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ 0 & a'_{3,2} & a'_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \end{Bmatrix}$$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a'_{2,2} & a'_{2,3} \\ 0 & 0 & a''_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \end{Bmatrix}$$

3.5 Systems of Linear Equations [Recall - Gaussian Elimination]

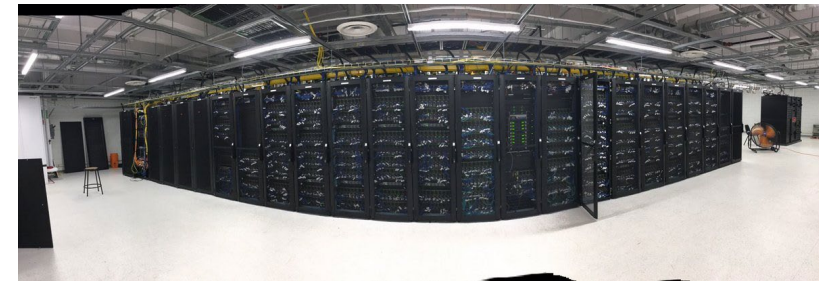
Benefits

- We achieve the 'exact' solution in one pass.
- Straightforward algorithm

Challenges

- Not very efficient (many calculations needed)
- Divide by zero is a possibility (if there is a zero on the diagonal)
- Round-off error

3.5 Systems of Linear Equations [FLOPS]



Floating-point operations per second (FLOPS) – a measure of computing performance

1 petaFLOPS is equal to 1,000,000,000,000,000 FLOPS ($\times 10^{15}$)

Waterloo Graham Cluster (33,000 CPU cores) ~2.6 petaFLOPS (retired)

Waterloo Nibi Cluster (140,000 CPU cores) ~15 petaFLOPS [186th globally in November 2025]

Fastest in the world (<https://www.top500.org/>) 1809 petaFLOPS

* On modern computers, the time required to do addition/subtraction and multiplication/division operations is similar, so we can add these up to understand the compute time required for an algorithm.

$$\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

1 multiplication, 1 subtraction, 1 division

Row#2 = Row#2 - (a_{21}/a_{11}) * Row#1

(1 multiplication, 1 subtraction) * (n-1)

Total # FLOPS per row:

$1 + (n-1) + 1 = n + 1$ multiplications

$1 + (n-1) = n$ subtractions

(forward elimination for one row)

3.5 Systems of Linear Equations [FLOPS]

Total number of flops per system of 'n' equations for:

1. Forward elimination
2. Backward substitution

Additional terms on order of n^2 , neglect for large values of n
 Section 9.2.1 gives a detailed description of cycle counting.

n	Forward elimination	Back substitution	Total flops	$\frac{2n^3}{3}$	% due to elimination
10	705	100	805	667	87.58
100	671550	10000	681550	666667	98.53
1000	6.67×10^8	1×10^6	6.67×10^8	6.67×10^8	99.85

$=705/805 * 100\%$

- 1) For large systems, as n increases the computation effort increases significantly.
- 2) Most of the effort is in the elimination step, emphasizes the need for more efficient direct methods.

3.6 Systems of Linear Equations [Gaussian Elimination-Pivoting]

- Pivoting – rearrange order of equations to avoid division by zero
 - Partial pivoting – moving rows so the **largest coefficients** are in the pivot (**diagonal**) positions
 - Full pivoting – moving both rows and columns (not commonly used – complicated to implement)

Pivot element $a_{11} = 0$

$$\begin{bmatrix} 0 & 1 & 2 \\ 0.01 & 3 & 3 \\ -10 & 7 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \\ 5 \end{Bmatrix}$$

Row#2 = Row#2 – (a_{21}/a_{11})*Row#1 -> divide by zero

Partial pivoting – interchange Row#1 with row that has the largest coefficient value (Row#3)

$$\begin{bmatrix} -10 & 7 & 1 \\ 0.01 & 3 & 3 \\ 0 & 1 & 2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 2 \\ 1 \end{Bmatrix}$$

Remember to pivot {b}

*** apply pivoting again to the diagonal element in the 2nd row when operating on rows 2 and 3.

3.6 Systems of Linear Equations [Gaussian Elimination-Pivoting]

• Example (problem 9.13b from the text)

9.13 Solve:

$$x_1 + x_2 - x_3 = -3$$

$$6x_1 + 2x_2 + 2x_3 = 2$$

$$-3x_1 + 4x_2 + x_3 = 1$$

with (a) naive Gauss elimination, (b) Gauss elimination with partial pivoting,



[A]				{B}	
1	1	-1	x1	-3	
6	2	2	x2	2	
-3	4	1	x3	1	
1	1	-1	x1	-3 R1	
0	-4	8	x2	20 R2' = R2 - R1*a21/a11	
-3	4	1	x3	1 R3	
1	1	-1	x1	-3	
0	-4	8	x2	20	
0	7	-2	x3	-8 R3' = R3 - R1*a31/a11	
1	1	-1	x1	-3	
0	-4	8	x2	20	
0	0	12	x3	27 R3'' = R3' - R2'*a32/a22	
Backward substitution					
			x3 =	2.25	
			x2 =	-0.5	
			x1 =	-0.25	

(b) The system is first expressed as an augmented matrix:

$$\begin{bmatrix} 1 & 1 & -1 & -3 \\ 6 & 2 & 2 & 2 \\ -3 & 4 & 1 & 1 \end{bmatrix}$$

We move row 2 to row 1, largest pivot element

Forward elimination: First, we pivot by switching rows 1 and 2:

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 1 & 1 & -1 & -3 \\ -3 & 4 & 1 & 1 \end{bmatrix}$$

Multiply row 1 by $1/6 = 0.16667$ and subtract from row 2 to eliminate a_{21} . Multiply row 1 by $-3/6 = -0.5$ and subtract from row 3 to eliminate a_{31} .

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 0 & 0.66667 & -1.33333 & -3.33333 \\ 0 & 5 & 2 & 2 \end{bmatrix}$$

Pivot:

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 0 & 5 & 2 & 2 \\ 0 & 0.66667 & -1.33333 & -3.33333 \end{bmatrix}$$

We move row 3 to row 2, largest pivot element

Multiply row 2 by $0.66667/5 = 0.133333$ and subtract from row 3 to eliminate a_{32} .

$$\begin{bmatrix} 6 & 2 & 2 & 2 \\ 0 & 5 & 2 & 2 \\ 0 & 0 & -1.6 & -3.6 \end{bmatrix}$$

Back substitution:

$$x_3 = \frac{-3.6}{-1.6} = 2.25$$

$$x_2 = \frac{2 - 2(2.25)}{5} = -0.5$$

$$x_1 = \frac{2 - 2(2.25) - 2(-0.5)}{6} = -0.25$$

3.6 Systems of Linear Equations [Gaussian Elimination-Pivoting+scaling]

- A well-conditioned system is one where a small change in the coefficients leads to a similar magnitude of change in the results.
- An ill-conditioned system occurs when small changes in the coefficients lead to large changes in the solution (or a wide range of answers can approximately satisfy the equations).

- Consider:

$$\begin{aligned}x_1 + 2x_2 &= 10 \\ 1.1x_1 + 2x_2 &= 10.4\end{aligned}$$



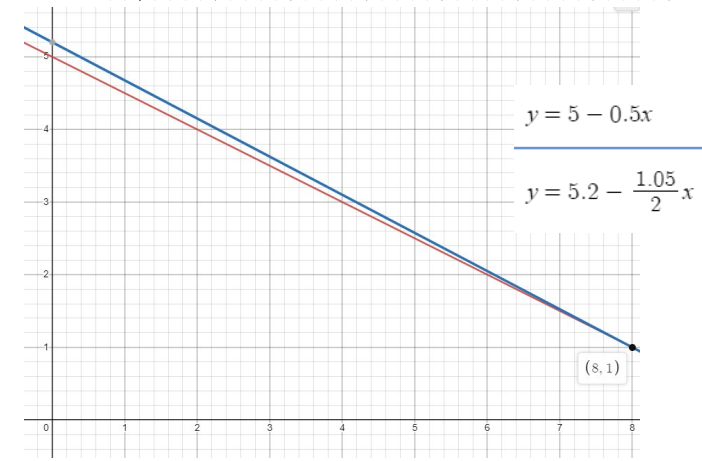
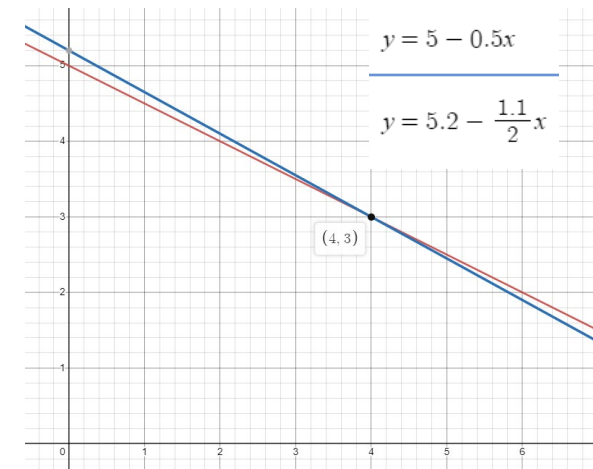
$$\begin{aligned}x_1 &= \frac{2(10) - 2(10.4)}{1(2) - 2(1.1)} = 4 \\ x_2 &= \frac{1(10.4) - 1.1(10)}{1(2) - 2(1.1)} = 3\end{aligned}$$

Change x_1 coefficient from 1.1 to 1.05

$$\begin{aligned}x_1 + 2x_2 &= 10 \\ 1.05x_1 + 2x_2 &= 10.4\end{aligned}$$



$$\begin{aligned}x_1 &= \frac{2(10) - 2(10.4)}{1(2) - 2(1.05)} = 8 \\ x_2 &= \frac{1(10.4) - 1.1(10)}{1(2) - 2(1.05)} = 1\end{aligned}$$



3.6 Systems of Linear Equations [Gaussian Elimination-Pivoting+Scaling]

- Partial pivoting
 - Avoid divide by zero issues (if any pivot elements are zero)
 - Also helps to avoid round-off errors that may be caused by small (non-zero) pivot element values.
 - * note that round-off errors can be cumulative in Gaussian elimination since each result depends on previous results (error accumulates). This is an important issue for large numbers of equations.
- Scaling
 - Scaling or normalizing the equations – dividing by the largest coefficient, so that the maximum coefficient is 1.0.

3.6 Systems of Linear Equations [Gaussian Elimination-Round-off Error]

- Example – solve using Gaussian elimination (carrying only 3 significant digits)

* this is unusual, but will be done here to make the point re roundoff

$$2x_1 + 100,000x_2 = 100,000$$

$$x_1 + x_2 = 2$$

Forward elimination gives:

$$2x_1 + 100,000x_2 = 100,000$$

$$-50,000x_2 = -50,000$$

$$x_2 = 1.0$$

$$x_1 = 0.0$$

If we didn't round off, this eqn would be:

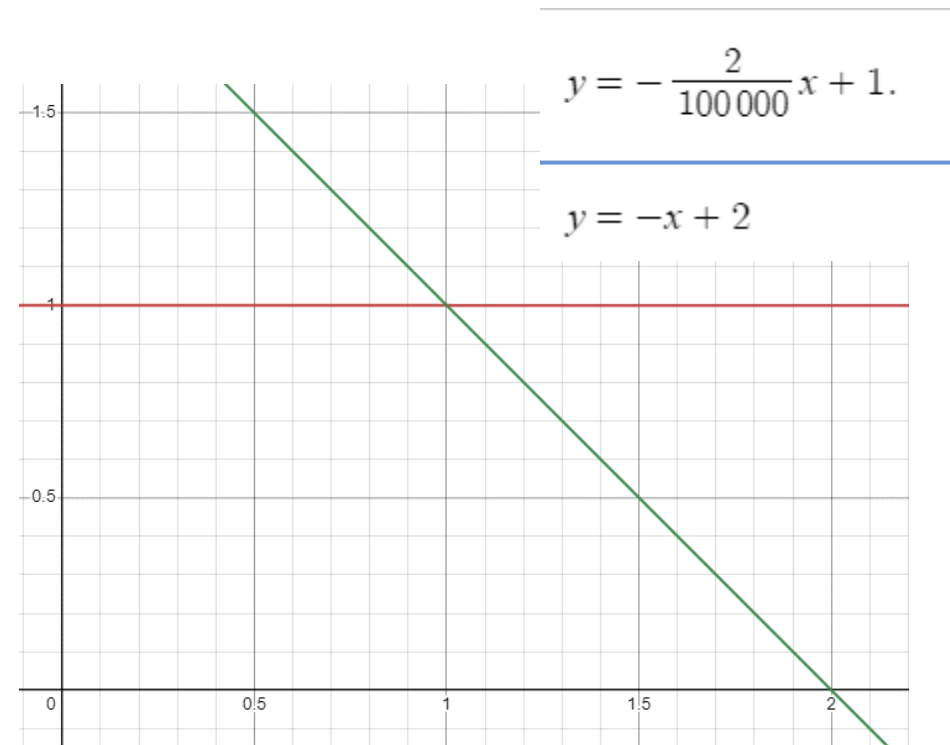
$$-49,999x_2 = -49,998$$

$$x_2 = 0.99998$$

$$x_1 = 1.00002$$

This is the true/accurate answer

No pivoting in this case - maximum coefficient is already in position a_{11}



3.6 Systems of Linear Equations [Gaussian Elimination-Round-off Error]

- Example – solve using Gaussian elimination (carrying only 3 significant digits)

* use scaling and partial pivoting this time

$$2x_1 + 100,000x_2 = 100,000$$

$$x_1 + x_2 = 2$$

Scaling:

$$0.00002x_1 + x_2 = 1$$

$$x_1 + x_2 = 2$$

Divide first eqn by 100,000

No scaling on 2nd eqn (max coefficient is 1.0)

Pivoting:

$$x_1 + x_2 = 2$$

$$0.00002x_1 + x_2 = 1$$

Next we need to pivot to get the largest coefficient into the a_{11} position

Forward elimination: $x_1 + x_2 = 2$

$$x_2 = 1.00 \text{ (3 sig figs)}$$

$$\text{Then } x_1 = 1.00$$

Even with rather aggressive rounding off, we achieve an answer accurate to 3 significant figures.

*** Figure 9.6 in Chapra provides pseudocode for Gauss elimination with partial pivoting